FOOD INVENTORY MANAGEMENT SYSTEM

Micro Project Report

Submitted to the APJ Abdul Kalam University, Kerala
for the partial fulfillment of the requirements for the award of the degree

of

Bachelor of Technology
in
Computer Science and Engineering

By

NANDANA MOHAN PRN25CS087
MALAVIKA S PRN25CS074
JITHENDU S PRN25CS064
PRADEESHA S PRASAD PRN25CS096

Guided by
ANISH A AZIZ , ASSISTANT PROFESSOR , COLLEGE OF ENGINEERING PERUMON

during
OCTOBER 2026



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
COLLEGE OF ENGINEERING PERUMON
CAPE - GOVT. OF KERALA, KOLLAM(D), KERALA

Table of Contents

Chapter No. Title

1. **Introduction**
2. **Scope of the project**
3. **Technologies used**
4. **Problem statement**
5. **Literature review or background**
6. **Implementation**
7. **Results and Discussion**
8. **Conclusion**
9. **References**

Introduction

RESQ (Emergency Resources and Rescue Management System) is a Java-based application developed to simplify emergency reporting and manage rescue operations effectively. Emergencies such as road accidents, fires, and other unexpected incidents require quick communication and proper coordination between the public and responsible authorities.

The system allows users to log in, report emergency incidents, view emergency details, and track the status of rescue operations. It also provides an administrative interface to allocate resources, manage available resources, and maintain the history of reported incidents.

The main aim of this project is to improve emergency response coordination, ensure proper resource utilization, and help authorities manage emergency situations more efficiently through a simple and user-friendly application.

Scope of the project

The scope of the RESCUE (Emergency Resources and Rescue Management System) is to provide an organized platform for reporting emergencies and managing rescue operations. The system allows users to report emergency incidents, view reported details, and track the progress of rescue activities.

It also provides administrators with the facility to allocate required resources, manage available resources, and maintain incident history. The application helps improve communication between users and authorities and supports better coordination during emergency situations.

In the future, the system can be enhanced with features such as real-time location tracking, GPS integration, instant notifications, and direct communication with emergency services. This will make emergency response faster and more effective.

Technology used

1. Java: Used as the main programming language to implement the application's logic and inventory management operations.

2. Java Swing: Used to develop the graphical user interface (GUI), including login screens, forms, tables, and the dashboard.

3. Object-Oriented Programming (OOP): Used to organize the application into classes and objects, making the code modular and easier to maintain.

4. File Handling: Used to store and retrieve inventory data locally between application sessions.

5. VS Code:Used as the integrated development environment (IDE) for writing, running, and debugging the Java application.

6. Git and GitHub: Used for version control, source code management, and collaboration among team members.

Problem statement

During emergency situations such as road accidents, fires, and other unexpected incidents, reporting emergencies and coordinating rescue operations can be difficult. Lack of proper communication, delays in reporting, and improper resource allocation may affect the response process. There is a need for a simple and organized system to report emergencies and manage rescue activities efficiently.

The RESCUE system aims to solve these problems by providing a platform where users can report emergency incidents, view incident details, and track the status of rescue operations. It also helps administrators allocate and manage resources and maintain incident history. The system is developed to improve communication, coordination, and emergency management.

Literature Review / Background

Traditional inventory management systems often rely on manual registers or spreadsheets to maintain stock records. These methods can be time-consuming and prone to errors, making it difficult to track quantities, monitor expiry dates, and identify low-stock items.

Modern inventory management applications improve efficiency by automating item recording, stock updates, searching, and report generation. Features such as expiry tracking and low-stock monitoring help reduce food wastage and prevent stock shortages.

Java provides object-oriented programming features for developing modular and maintainable applications, while Java Swing enables the creation of graphical user interfaces. Local file handling allows inventory data to be stored and retrieved without requiring a separate database system.

The Food Inventory System applies these technologies to provide a simple solution for managing food items, organizing categories, monitoring stock, tracking expiry dates, and generating inventory reports. It offers a practical approach to basic inventory management while demonstrating the application of Java programming and object-oriented concepts.

Implementation

MODULES OF THE FOOD INVENTORY SYSTEM

The Food Inventory System consists of several modules that work together to manage food items, monitor stock levels, track expiry dates, and maintain inventory records efficiently.

1. Login Module

This module provides secure access to the application through a login interface. It validates the username and password and allows access to the dashboard when the entered credentials are correct.

2. Food Item Management Module

This module allows users to add, update, view, and delete food items. Each record contains details such as food ID, food name, category, quantity, unit, price, and expiry date. It also validates the entered data to maintain accurate inventory records.

3. Inventory Management and Search Module

This module displays all available food items in a tabular format. Users can search for items by food ID, name, or category, view their details, and select records for modification or deletion.

4. Category Management Module

This module organizes food items into categories, making inventory records easier to manage and locate. Users can add new categories and manage existing ones.

5. Stock Management Module

This module allows users to increase or decrease the quantity of food items based on stock received or consumed. It helps maintain accurate quantity records and prevents stock from being reduced below the available quantity.

6. Expiry and Low-Stock Monitoring Module

This module identifies expired food items, items approaching their expiry dates, and items with low stock levels. It helps users take timely action to reduce food wastage and avoid stock shortages.

7. Reports Module

This module provides inventory summaries and detailed reports, including total food items, low-stock items, expired items, and items expiring soon. It helps users understand the current inventory status.

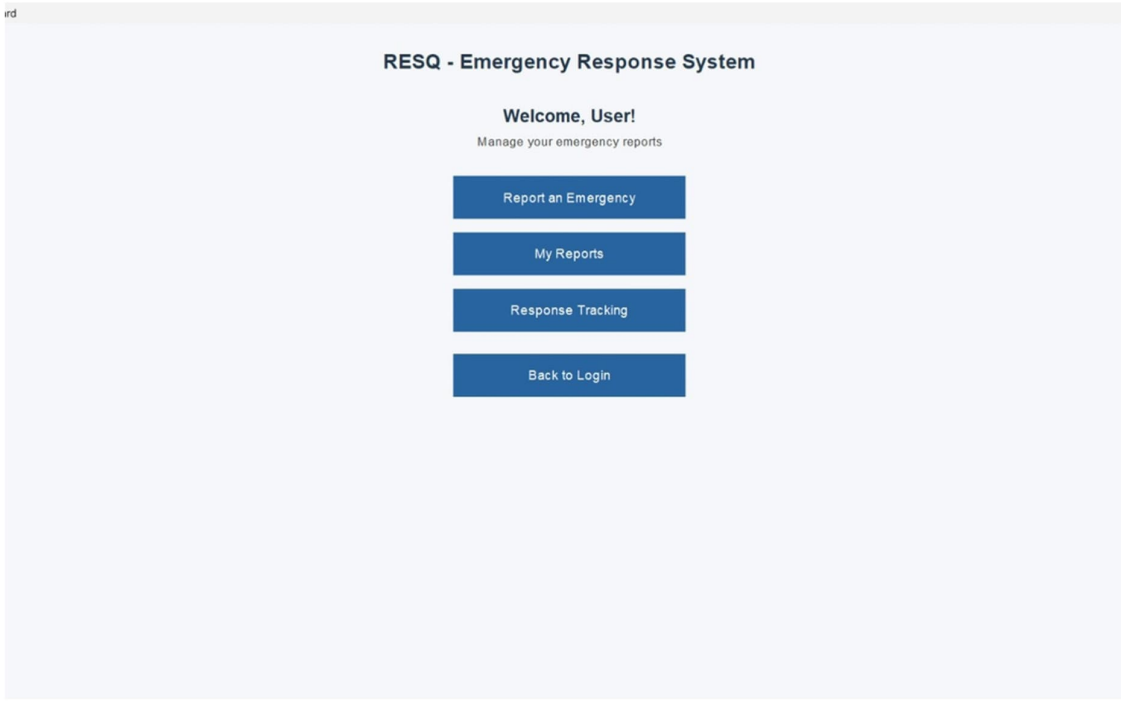
8. Data Storage and Validation Module

This module manages the saving and retrieval of inventory data using local file storage. It validates user inputs, handles errors, and preserves inventory records between application sessions.

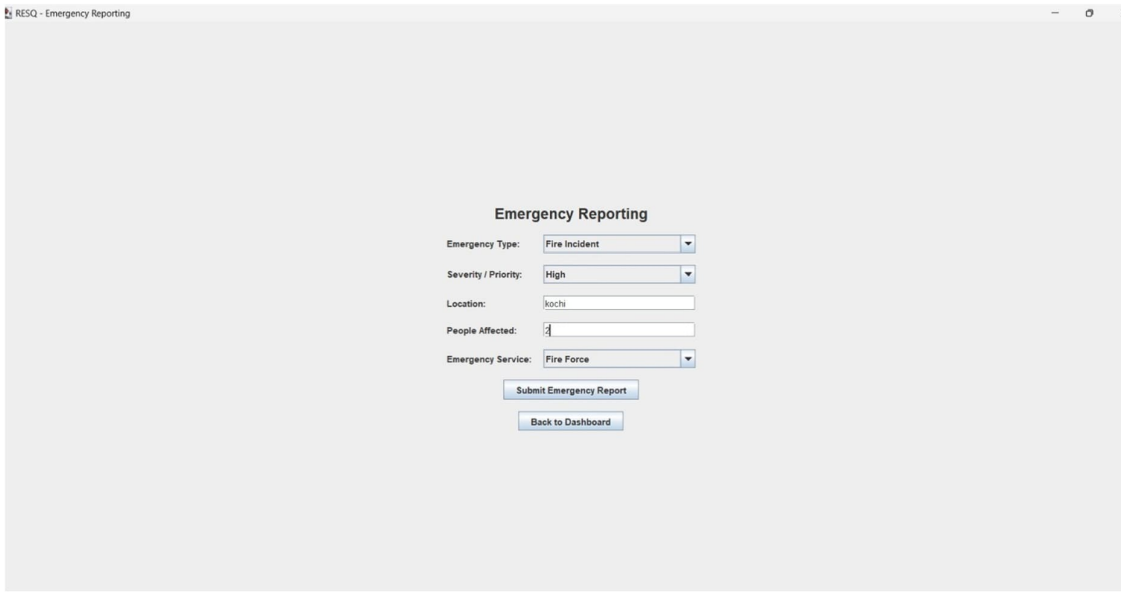
9. Dashboard Module

The dashboard serves as the main interface of the application. It displays important inventory statistics and provides navigation to the different modules, allowing users to access the system's features conveniently.

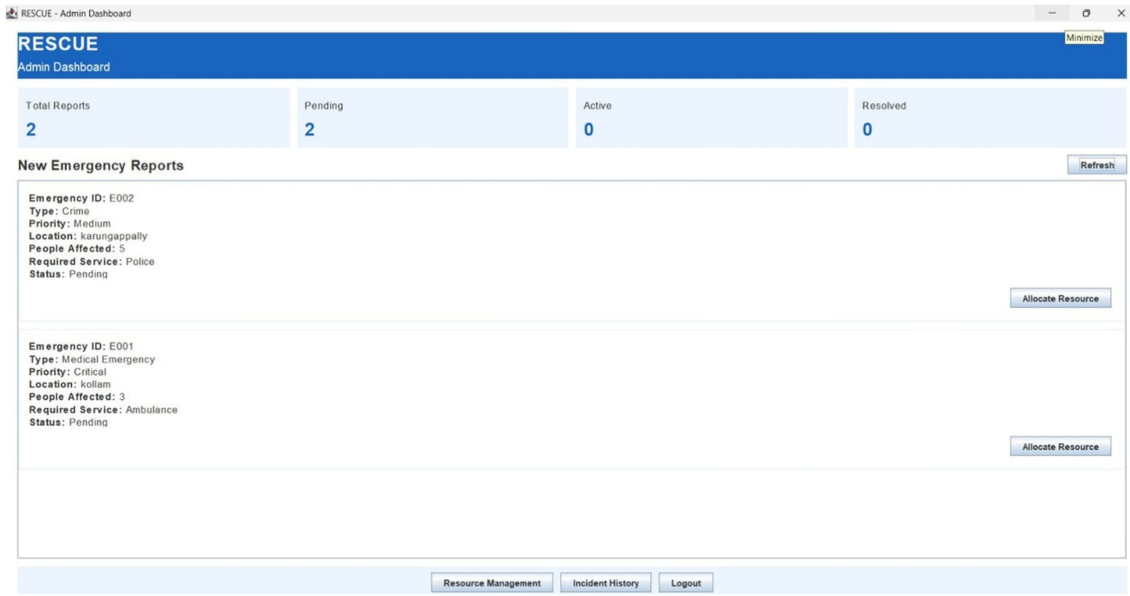
User Dashboard Frame



Emergency Reporting



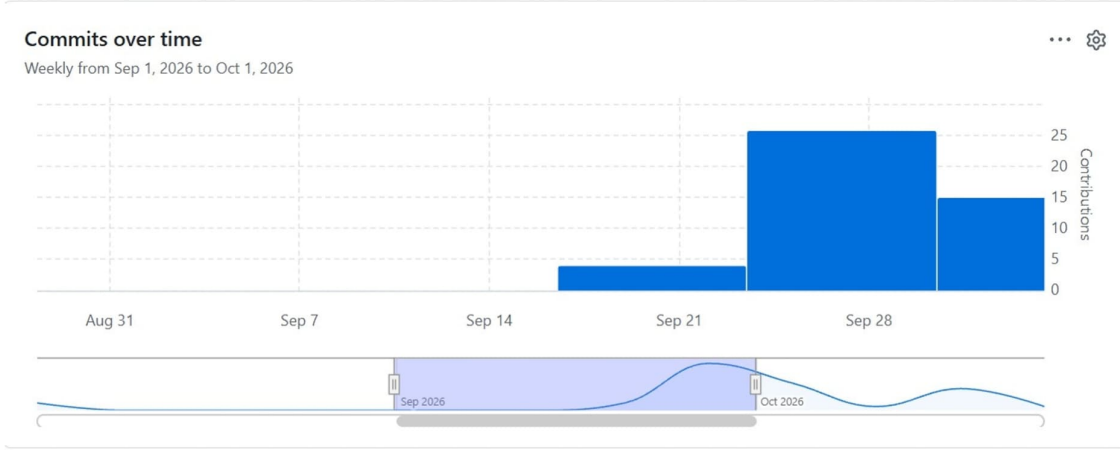
Admin Dashboard

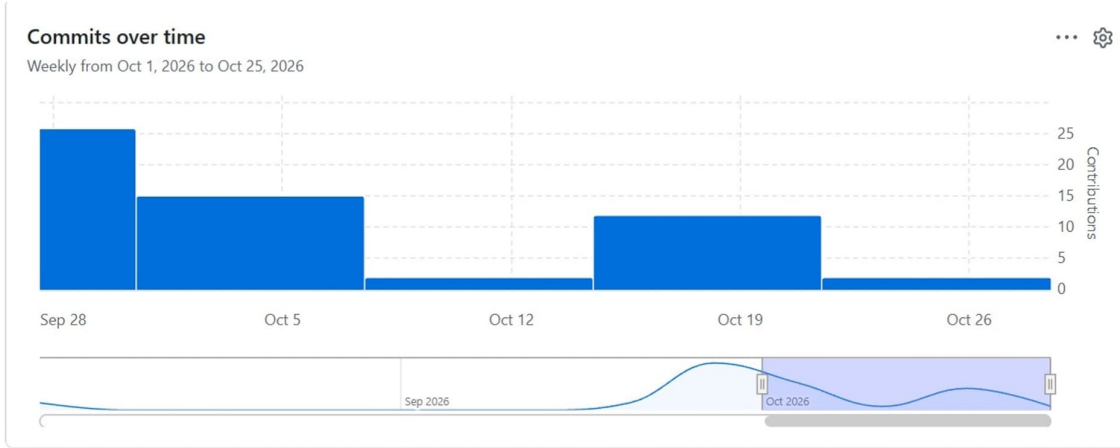


Stock Management Frame

Results and Discussion

The Food Inventory System was successfully developed using Java and Java Swing. It allows users to manage food items, update stock, organize categories, and track expiry dates and low-stock items. The dashboard and reports provide useful inventory information, while local file storage maintains records between sessions. Overall, the system simplifies inventory management, reduces manual effort, and demonstrates the practical application of object-oriented programming and file handling in Java.





Conclusion

The Food Inventory System provides an efficient way to manage food items, monitor stock levels, track expiry dates, and maintain inventory records. Features such as item management, low-stock alerts, and reports reduce manual effort and improve record-keeping accuracy. The application uses Java Swing for its graphical interface and local file storage to preserve data.

The project demonstrates key OOP concepts, including encapsulation, which combines data and methods within classes, and abstraction, which simplifies inventory operations. Inheritance and polymorphism are applicable where implemented in the code. Classes and objects organize the system, while exception handling helps manage errors and invalid inputs.

Overall, the project provides practical experience in Java programming, GUI development, file handling, and object-oriented design.

GitHub path

https://github.com/nandanamohan-tech/OOP/tree/main/src/FoodInventorySystem

Reference

1. Oracle, Java Documentation. https://docs.oracle.com/en/java/

2. Oracle, The Java Swing Tutorial. https://docs.oracle.com/javase/tutorial/uiswing/

3. Oracle, Java SE API Documentation. https://docs.oracle.com/en/java/javase/25/docs/api/

4. GitHub, Food Inventory System – Project Source Code. https://github.com/nandanamohan-tech/OOP/tree/main/src/FoodInventorySystem